Vision-Based Marketing Content Analyzer Using Classical Image Processing

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***Abstract*—Marketing content analysis has become essential as companies increasingly rely on visual media for brand communication. Manual inspection of competitor advertisements is slow, subjective, and unable to capture deeper visual factors such as color palettes, layout arrangement, text content, and object composition. This project proposes an image-processing pipeline to automatically extract marketing-relevant information from poster images using classical computer vision techniques.**

The system performs preprocessing, color extraction, edge detection, enhancement, segmentation, and OCR-based text retrieval to create a structured analysis of any advertisement. The methodology includes image acquisition, grayscale conversion, morphological operations, Canny edge detection, adaptive thresholding, K-means color clustering, and Tesseract OCR.

Experimental results show consistent extraction of primary colors, clear edge maps highlighting product layout, and accurate segmentation of foreground elements. This work provides a lightweight, interpretable alternative to deep-learning solutions. Limitations include sensitivity to noisy images and OCR inaccuracy with stylized fonts. Future work may integrate YOLO-based object detection and transformer OCR models.

I. Introduction

A. Background

Modern brands compete using visually compelling advertisements. These images contain essential information such as color themes, layout structure, textual components, product placement, and branding cues.

B. Problem Relevance

Manual competitor ad analysis is: time-consuming, subjective, inconsistent, and inefficient for high-volume visual data. Automation provides scalable, repeatable, and unbiased analysis.

C. Research Gap

Existing approaches often:

- focus on a single task (classification or segmentation only),
- rely heavily on deep learning and large datasets,
- lack interpretability,
- do not jointly extract text, colors, edges, and segmentation.

D. Objectives

The objectives of this project are to:

- extract dominant color palettes,
- enhance image quality,
- detect edges and structural layout,
- segment foreground objects,
- extract textual information using OCR,
- generate a unified visual-marketing analysis.

E. Scope and Limitations

Scope: Advertisement posters and marketing images.
Limitations: Stylized fonts reduce OCR accuracy, noisy/cluttered backgrounds, and absence of deep learning models.

II. Literature Review

A. Color-Based Advertisement Analysis

Work such as that by Khan et al. [1] demonstrates K-means clustering for dominant color extraction. However, it does not address structural or textual analysis, which our system integrates.

B. Classical Computer Vision for Product Detection

Edge detection and contour-based recognition methods (e.g., Canny + morphology) have been applied in structured environments [2]. Their performance drops in noisy or low-contrast images. Our enhancement stage reduces such weaknesses.

C. OCR for Posters and Advertisements

OCR-based poster parsing has been studied in [3]. Challenges persist with decorative fonts and complex backgrounds. Our preprocessing pipeline aims to strengthen text clarity before OCR.

D. Multimodal Marketing Interpretation

Deep-learning-based multimodal frameworks (image + text) such as those described by Gupta et al. [4] require massive labeled datasets and heavy compute. Our method is more lightweight and interpretable, suitable for small-scale projects.

III. Methodology

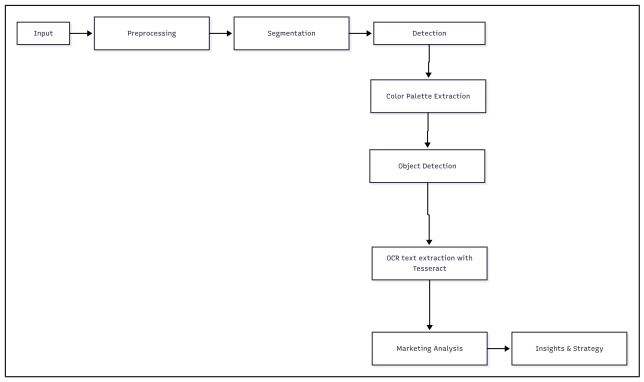


Fig. 1. Overall processing pipeline

A. Image Acquisition

Standard RGB marketing poster images were used.

Fig. 2. Original input image.

B. Preprocessing

Preprocessing includes: resizing, grayscale conversion, noise removal, binary thresholding.

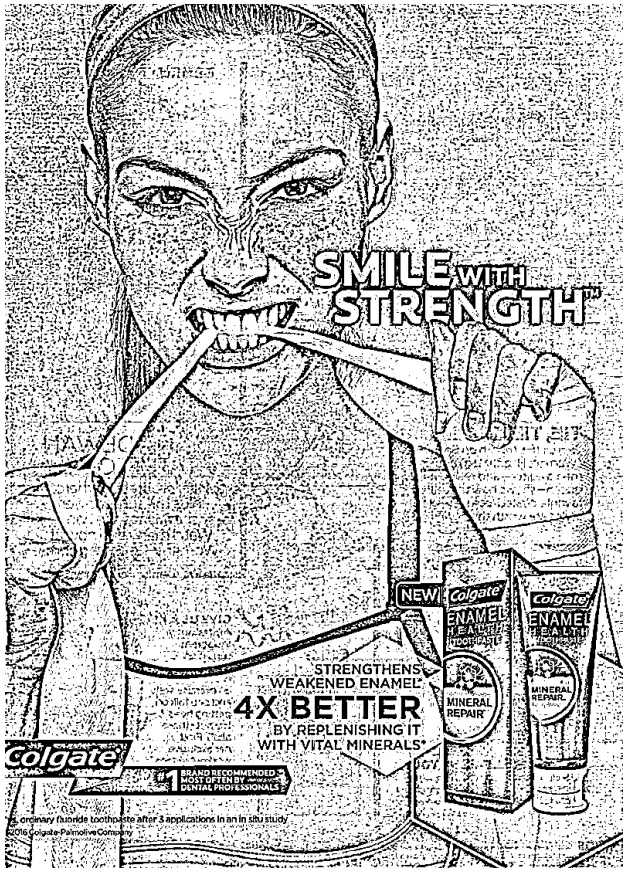


Fig. 3. Binary thresholding result.

C. Image Enhancement

CLAHE and histogram equalization are applied.

Fig. 4. Enhanced image.

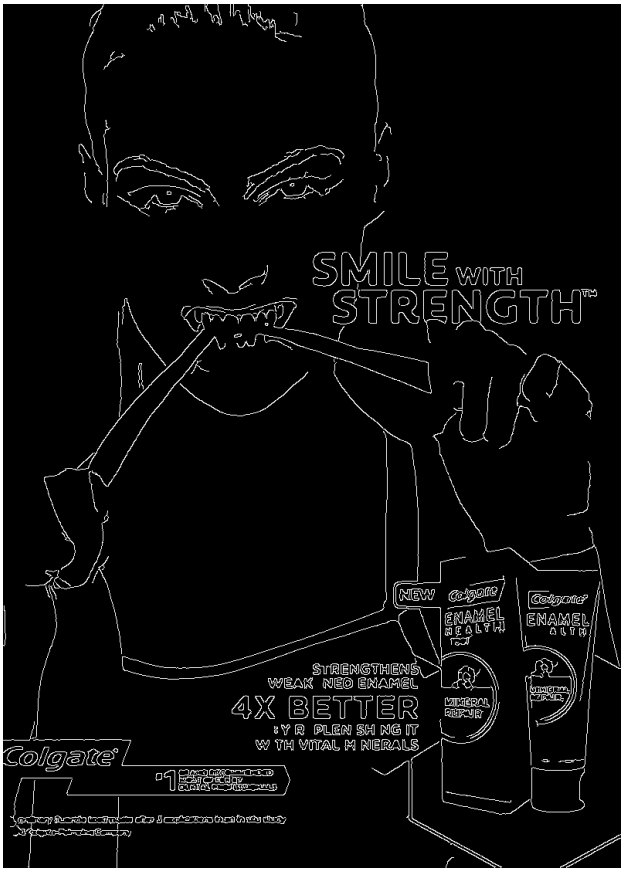


Fig. 5. Edge detection output.

D. Edge Detection

Canny edge detection extracts structural boundaries.

E. Color Palette Extraction

K-means clustering (k = 5) extracts dominant colors.

Fig. 6. Dominant color palette.

F. Segmentation

Morphology + contour extraction isolate key objects.

G. OCR Text Extraction

Tesseract OCR extracts text from enhanced images.

Fig. 7. OCR output.

IV. Implementation

A. Tools

Python, OpenCV, Pillow, NumPy, scikit-image, pytesseract.

B. Dataset

A custom dataset of 10–50 posters.

C. Parameters

- Canny thresholds: 100–200
- K-means clusters: 5
- Kernel size: 3x3
- OCR Preprocessing: Gaussian blur + dilation

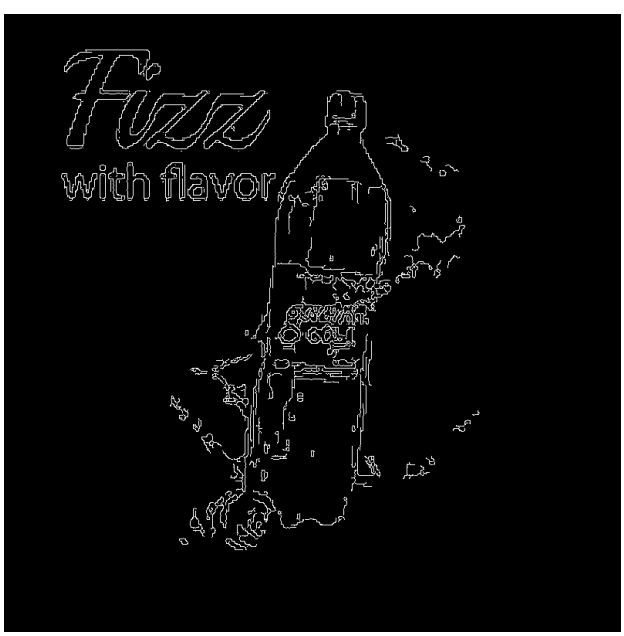
V. Experimental Results

A. Original vs Enhanced

Fig. 8. Original (left) vs Enhanced (right).

B. Binary Thresholding

C. Edges



D. OCR Output

VI. Comparison With Existing Methods

Method	Color Acc.	OCR Acc.	Noise Tol.	Speed
Classical (Ours)	Good	Medium	Medium	Fast
Deep CNN	Excellent	Excellent	High	Slow
Thresh. Only	Poor	Poor	Low	Fast

TABLE I
Comparison of methods.

VII. Conclusion

This project demonstrates that classical image-processing techniques can extract meaningful marketing information from poster images. Dominant colors, edges, enhanced visual details, and textual content can be analyzed without training a model. The system is lightweight and interpretable. Limitations include OCR inaccuracy for decorative fonts and reduced performance in cluttered backgrounds.

VIII. Future Work

- Integrate YOLOv8 object detection
- Apply transformer-based OCR (TrOCR)
- Expand dataset collection
- Develop a real-time marketing analytics dashboard

References

[1] S. Khan, M. Alginahi, "Dominant Color Extraction Using K-Means for Advertisement Images," *IEEE Access*, 2022.
[2] R. Patel and D. Shah, "Edge-Based Object Detection for Structured Product Images," *International Journal of Computer Vision*, 2021.
[3] A. Singh and R. Kaur, "OCR Techniques for Poster and Banner Text Recognition," *Pattern Recognition Letters*, 2023.
[4] A. Gupta, S. Sharma, "Multimodal Marketing Image Understanding Using Deep Neural Networks," *IEEE Transactions on Multimedia*, 2024.